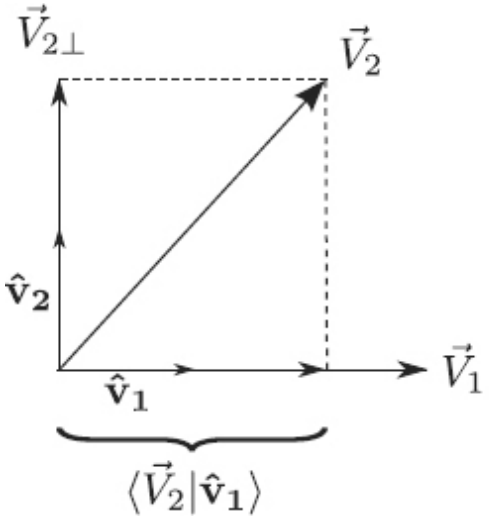
The final part of the theorem states that the eigenvectors are complete. In other words, if the space is *N*-dimensional, there will be *N* orthonormal eigenvectors. The proof is easy and I will leave it to you.

> **Exercise 3.1:** Prove the following: If a vector space is *N*-dimensional, an orthonormal basis of *N* vectors can be constructed from the eigenvectors of a Hermitian operator.

3.1.6 The Gram-Schmidt Procedure

Sometimes we encounter a set of linearly independent eigenvectors that *do not* form an orthonormal set. This typically happens when a system has degenerate states—distinct states that have the same eigenvalue. In that situation, we can always use the linearly independent vectors we have, to create an orthonormal set that spans the same space. The method is the Gram-Schmidt procedure I alluded to earlier. Fig. 3.1 illustrates how it works for the simple case of two linearly independent vectors. We start with the two vectors $\vec{V}_1$ and $\vec{V}_2$, and from these we construct two orthonormal vectors, $\hat{\mathbf{v}}_1$ and $\hat{\mathbf{v}}_2$.



$$\hat{\mathbf{v}}_1 = \frac{\vec{V}_1}{|\vec{V}_1|}$$

$$\vec{V}_{2\perp} = \vec{V}_2 - \langle \vec{V}_2 | \hat{\mathbf{v}}_1 \rangle$$

$$\hat{\mathbf{v}}_2 = \frac{\vec{V}_{2\perp}}{|\vec{V}_{2\perp}|}$$